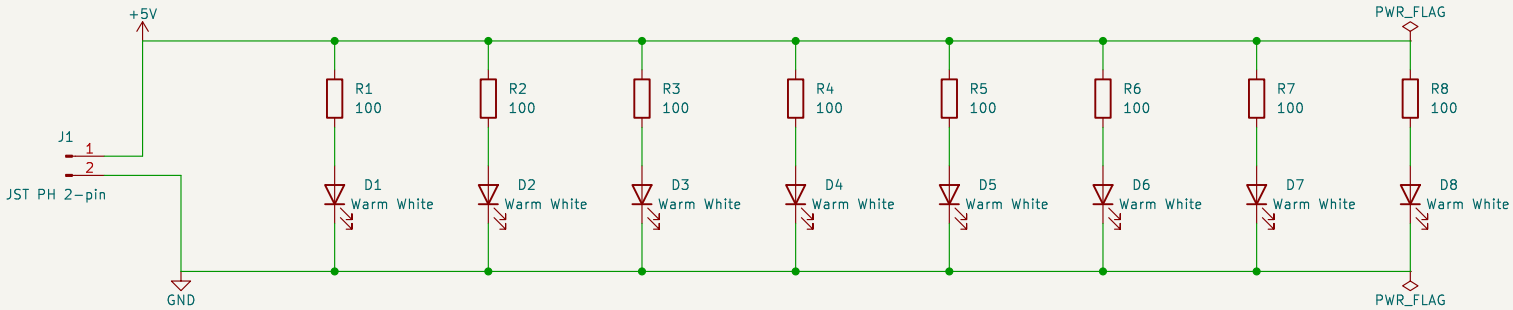
8 LEDs, each with its own resistor (parallel branches).
I(LED) ≈ (5.0 V − 2.9 V) / 100 Ω ≈ 21 mA (typical LED-strip drive current)
Total ≈ 168 mA from USB. P(R) ≈ 44 mW (0805 rating 125 mW).
D1–D8: XINGLIGHT XL-2835WWC-02 warm white 2835 (2800–3000 K), LCSC C2843875
R1–R8: UNI-ROYAL 0805W8F1000T5E 100R 1% 0805, LCSC C17408



J1: JST PH 2.0 mm right-angle socket (S2B-PH-SM4-TB).
Pin 1 = +5V, pin 2 = GND — check the cable before plugging in.

Assembled by JLCPCB (economic PCBA, top side)
8x warm white 2835 LED light bar, 100 x 15 mm, JST PH 2.0 power input
Banyule Nillumbik Tech School

Sheet: /		
File: usb-led-bar-jst.kicad_sch		
Title: USB LED Bar		
Size: A4	Date: 2026-10-15	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1